Student-Submitted Questions & Answers
Part 1

Exam III Review Day

Elementary Linear Algebra (MATH 2250-03)

Spring, 2019

Introduction

In this document, I have compiled all of the questions that were sent to me in preparation for the Exam 3 Review Day. These questions are near-verbatim the questions I was sent (minus any personally identifying information and with minor formatting changes to make the math display nicely). After each question, I've provided a short answer. Hopefully, seeing the questions your classmates asked (phrased exactly as they asked them) will provide a helpful 'on-the-ground' perspective regarding the types of questions you might have and the answers to them.

Questions

1. Here is my question, going the other direction from the if and only if statement in Theorem 14 of the book:

   Let A be an $m \times n$ matrix whose columns are linearly independent. Show that $A^T A$ is invertible.

   This is a good, challenging problem! Very nice! We'll give the proof in a second, but first, let's prove a nice little supporting lemma.

   (By the way, just because most teachers never explain this to students this for some reason, the term 'lemma' is derived from the Attic Greek word for 'gift' or 'bribe' and is just a name mathematicians use for little 'helping theorems' that let us prove a larger result. So, lemmas are just theorems, but we call them lemmas to let the reader know that they're not the thing we're really trying to prove.)

   Lemma 1. *Let W be a subspace of $\mathbb{R}^n$ and suppose $x \in \mathbb{R}^n$ is a vector which satisfies both $x \in W$ and $x \in W^\perp$. Then, $x = 0$.*

   Proof. By definition, $W^\perp$ is the set of all vectors v which satisfy $v \cdot w = 0$ for every vector $w \in W$. So, since $x \in W^\perp$ and $x \in W$, we have $x \cdot x = 0$. But notice that this then implies that $x \cdot x = ||x||^2 = 0$, which must mean that $x = 0$. □

   So, with this in hand, let's prove the original theorem now:

   Proof. Let A be an $m \times n$ matrix whose columns are linearly independent. Note that any solution x to the homogeneous system $A^T A x = 0$ must satisfy $Ax \in \mathrm{Nul}(A^T)$, as $A^T A x = A^T\big(Ax\big)$. Recalling that $\mathrm{Col}(A)^\perp = \mathrm{Nul}(A^T)$, then as $Ax \in \mathrm{Col}(A)$, any solution x to $A^T A x = 0$ must satisfy both $Ax \in \mathrm{Col}(A)$ and $Ax \in \mathrm{Col}(A)^\perp$. By the above lemma, this tells us that $Ax = 0$. So, any solution x to $A^T A x = 0$ must also be a solution to $Ax = 0$. But, as Ax is a linear combination of the columns of A, linear independence yields that the homogeneous system $Ax = 0$ has only the trivial solution $x = 0$. Hence, the homogenous system $A^T A x = 0$ has only the trivial solution $x = 0$, and as $A^T A$ is a square matrix, the invertibility of $A^T A$ follows immediately from the invertible matrix theorem. □

2. Can you give us a refresher on how to go about row reduction?
Check out these videos (the links should be clickable):
Row Reducing a Matrix - Systems of Linear Equations - Part 1
Row Reducing a Matrix - Systems of Linear Equations - Part 2

3. Problem 1 on Homework Assignment 9: can we go over that process and theory again? This feels really complicated and I don't think it is meant to be.
Problem 1 on Homework 9 is:

> Let $\mathrm{T} : \mathbb{P}_2 \to \mathbb{P}_1$ be the transformation that maps a polynomial, $p(t)$, to its derivative, $p'(t)$. Show that T is a linear transformation. Then, find the matrix for T relative to the basis $\mathcal{B} = \left\{1, t, t^2\right\}$ for $\mathbb{P}_2$ and the basis $\mathcal{D} = \left\{1, t\right\}$ for $\mathbb{P}_1$.

To answer your question, you're definitely correct in your intuition that solving this problem doesn't need to be a complicated process. To show this, let's just go through the solution to this problem given in the key and give some commentary and additional explanation. Below, I've posted the solution given in the key in red, and I've added addition explanation and detail in blue.

Solution: Before beginning, let's figure out what we want to find here:

(a) The problem asks us to show that T is linear. That is, we need to show that

$$\mathrm{T}\Big(cp(t)\Big) = c\mathrm{T}\Big(p(t)\Big) \quad \text{and} \quad \mathrm{T}\Big(p(t) + q(t)\Big) = \mathrm{T}\Big(p(t)\Big) + \mathrm{T}\Big(q(t)\Big)$$

for any polynomials $p(t)$ and $q(t)$ in $\mathbb{P}_2$ and any scalar $c \in \mathbb{R}$.

(b) The problem asks for the matrix for T relative to the basis $\mathcal{B} = \left\{1, t, t^2\right\}$ for $\mathbb{P}_2$ and the basis $\mathcal{D} = \left\{1, t\right\}$ for $\mathbb{P}_1$. That is, we need to find the 2×3 matrix M such that

$$\mathrm{M}[p(t)]_{\mathcal{B}} = \Big[\mathrm{T}\Big(p(t)\Big)\Big]_{\mathcal{D}}.$$

First, let's start by noticing that T is a linear transformation, since the derivative can always be split up across a sum and since constant multiples can always be pulled outside of a derivative. That is, since

$$\frac{d}{dt}\Big(c_1 f(t) + c_2 g(t)\Big) = c_1\left(\frac{d}{dt} f(t)\right) + c_2\left(\frac{d}{dt} g(t)\right)$$

for any functions $f(t)$ and $g(t)$ and any constants c_1 and c_2, any transformation which sends a function to its derivative is automatically a linear transformation. Since this is exactly what T does (i.e. T sends a polynomial function of degree at most 2 to its derivative, which is a polynomial function of degree at most 1), we may certainly conclude that T is a linear transformation.

So, since we know T is a linear transformation, let's start by seeing what T does to an arbitrary (or 'generic') polynomial in $\mathbb{P}_2$:
As $\mathrm{T} : \mathbb{P}_2 \to \mathbb{P}_1$ takes a polynomial $p(t) \in \mathbb{P}_2$ to its derivative $p'(t) \in \mathbb{P}_1$, let's see what T does to a generic polynomial $p(t) = a + bt + ct^2 \in \mathbb{P}_2$ where $a, b, c \in \mathbb{R}$ are arbitrary constants. That is, let's observe that

$$\mathrm{T}\Big(p(t)\Big) = \frac{d}{dt}\Big(a + bt + ct^2\Big) = b + 2ct.$$

Here, we just noticed that every polynomial in $\mathbb{P}_2$ can be written in the form $a + bt + ct^2$ for some constants $a, b, c \in \mathbb{R}$. For example, the polynomial $1 - 3t^2$ has $a = 1$, $b = 0$, and $c = -3$. So, what we're really doing here is seeing what T does to *any* polynomial in $\mathbb{P}_2$ by just replacing specific numbers with variables.

Even better, since the values of a, b, and c uniquely determine a given polynomial in $\mathbb{P}_2$, we might as well just talk about the values of these three constants. That is to say, this will all probably be much easier if we just deal with some vectors of the form $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ in $\mathbb{R}^3$ instead of polynomials of the form $a + bt + ct^2$ in $\mathbb{P}_2$. To do this, we'll just convert a polynomial $p(t)$ to its coordinate vector $[p(t)]_{\mathcal{B}}$, where $\mathcal{B}$ is the standard basis $\mathcal{B} = \{1, t, t^2\}$ of $\mathbb{P}_2$:

So, using the basis $\mathcal{B} = \{1, t, t^2\}$ for $\mathbb{P}_2$ and the basis $\mathbb{D} = \{1, t\}$ for $\mathbb{P}_1$, we have that the matrix for T relative to the bases $\mathcal{B}$ and $\mathcal{D}$ is the matrix M such that if

$$[p(t)]_{\mathcal{B}} = [a + bt + ct^2]_{\mathcal{B}} = \begin{bmatrix} a \\ b \\ c \end{bmatrix},$$

then

$$\mathrm{M}[p(t)]_{\mathcal{B}} = \left[\mathrm{T}\big(p(t)\big)\right]_{\mathcal{D}} = [b + 2ct]_{\mathcal{D}} = \begin{bmatrix} b \\ 2c \end{bmatrix}.$$

Here, we're just noticing that

$$\left[a + bt + ct^2\right]_{\mathcal{B}} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

(just like we mentioned would be convenient) and that

$$\left[\mathrm{T}\big(a + bt + ct^2\big)\right]_{\mathcal{D}} = [b + 2ct]_{\mathcal{D}} = \begin{bmatrix} b \\ 2c \end{bmatrix}.$$

So, finding the matrix for T relative to the basis $\mathcal{B}$ for $\mathbb{P}_1$ and the basis $\mathcal{D}$ for $\mathbb{P}_2$, all we have to do is find a 2×3 matrix $\mathrm{M} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \end{bmatrix}$ so that

$$\begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ 2c \end{bmatrix}.$$

Of course, we *could* do this by just directly solving for the values of the entries of M so that the matrix-vector multiplication above works out how we want it to. That is, we could compute

$$\begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} m_{11}a + m_{12}b + m_{13}c \\ m_{21}a + m_{22}b + m_{23}c \end{bmatrix}$$

and we could notice that setting $m_{11} = m_{13} = m_{21} = m_{22} = 0$ and setting $m_{12} = 1$, $m_{23} = 2$ gives

$$\begin{bmatrix} m_{11}a + m_{12}b + m_{13}c \\ m_{21}a + m_{22}b + m_{33}c \end{bmatrix} = \begin{bmatrix} (0)a + (1)b + (0)c \\ (0)a + (0)b + (2)c \end{bmatrix} = \begin{bmatrix} b \\ 2c \end{bmatrix}.$$

This does indeed produce the answer we want. That is, this tells us that the matrix for T relative to the basis $\mathcal{B}$ for $\mathbb{P}_1$ and the basis $\mathcal{D}$ for $\mathbb{P}_2$ is just the matrix $\mathrm{M} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

Of course, while this is *one way* to solve this problem, it's probably not a particularly *good way* to solve problems like this in general – or, at least, it definitely isn't an *efficient* way to solve problems like this. Quite naturally, we would probably want a better way to do this if we were, for example, solving this problem in the case where T sends $p(t) \in \mathbb{P}_{5000}$ to $p'(t) \in \mathbb{P}_{4999}$ instead, since the method we use above would require that we solve a system of 4999 equations involving $(5000)(4999) = 24995000$ variables.

In the answer provided in the key, we use a much more efficient method:
Using the formula for M provided in Section 5.4 of the textbook, we have

$$\begin{aligned} M &= \begin{bmatrix} [T(b_1)]_{\mathcal{D}} & [T(b_2)]_{\mathcal{D}} & [T(b_3)]_{\mathcal{D}} \end{bmatrix} \\ &= \begin{bmatrix} [T(1)]_{\mathcal{D}} & [T(t)]_{\mathcal{D}} & [T(t^2)]_{\mathcal{D}} \end{bmatrix} \\ &= \begin{bmatrix} [0]_{\mathcal{D}} & [1]_{\mathcal{D}} & [2t]_{\mathcal{D}} \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}. \end{aligned}$$

So, as we can see, this works! That is, we get the same answer as before, but *we didn't have to solve any systems of equations*! This is great (and definitely much faster). We can even check to see that this is the correct answer as follows:

Let's check that this actually works by confirming that $M[p(t)]_{\mathcal{B}} = \Big[T\big(p(t)\big)\Big]_{\mathcal{D}}$ for an arbitrary polynomial $p(t) = a + bt + ct^2 \in \mathbb{P}_2$:

$$\begin{aligned} M[p(t)]_{\mathcal{B}} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \\ &= \begin{bmatrix} b \\ 2c \end{bmatrix} \\ &= \Big[T\big(p(t)\big)\Big]_{\mathcal{D}}. \end{aligned}$$

So, this agrees with our previous calculation, and we may conclude that the matrix M given above is indeed the matrix for T relative to the bases $\mathcal{B}$ and $\mathcal{D}$.

4. In question 6 on homework 10, we show that $x \mapsto \text{proj}_L(x)$ is a linear transformation, where $L = \text{span}\{u\}$ is the one-dimensional subspace spanned by some vector u. Is the inverse of this true? That is, is the mapping $\text{proj}_L(x) \mapsto x$ is also always a linear transformation? If so, why?
This is a good question, and touches on something that we often take for granted in this class. Let's investigate and see what happens.

To begin, let's notice that if we have a vector space V and a vector space W, a linear transformation $T : V \to W$ is a **function** that sends $v \in V$ to $T(v) \in W$ while satisfying $T(cv) = cT(v)$ and $T(v + w) = T(v) + T(w)$ for all scalars c and all vectors $v, w \in V$.

With this in mind, let's recall that the mapping $T : \mathbb{R}^n \to L$ which sends $x \in \mathbb{R}^n$ to $\text{proj}_L(x) \in L$ is a linear transformation from $\mathbb{R}^n$ to L (like you said). But, let's notice that it might be the case that $T(x) = T(y)$ for some $x \neq y$. That is, since $T(x) = \frac{x \cdot u}{u \cdot u} u$, then if there exists a vector $w \in L^{\perp}$ such that $w \neq 0$, we have

$$\begin{aligned} T(x + w) &= \frac{(x + w) \cdot u}{u \cdot u} u \\ &= \frac{x \cdot u}{u \cdot u} u + \frac{w \cdot u}{u \cdot u} u \\ &= \frac{x \cdot u}{u \cdot u} u + (0)u = T(x). \end{aligned}$$

Then, if we wanted to send $T(x) = \text{proj}_L(x)$ to x, we would certainly also have to send $T(x + w) = \text{proj}_L(x + w)$ to $x + w$. But, since $T(x) = T(x + w)$, we would then have to send both $T(x)$ to *both x and $x + w$*! So, we would **no longer have a function** if we did this. That is, if $L^{\perp}$ is not the zero subspace, the mapping $\text{proj}_L(x) \mapsto x$ is **not a linear transformation** not because it fails to be linear or anything like that, but because it ***is not a function***.
However, if $L^{\perp} = \{0\}$, then this mapping *is* a linear transformation. But, in this case, we would need $\mathbb{R}^n = \mathbb{R}$ and $L = \mathbb{R}$ as well. So, unfortunately, the answer to this question is 'yes' only in the relatively boring case where $\text{proj}_L(x) = x$ for every $x \in \mathbb{R}^n$, which only happens when $n = 1$ and when the vector u is just a nonzero number $u \in \mathbb{R}$.

5. What is the use of making a orthonormal basis for a space out of the column space of a given basis, if you know the column and null space of the basis, wouldn't an appropriately sized unit vector from of a modified identify matrix accomplish much of the same? What's the benefit of transforming a basis into a orthonormal basis, as far as retaining any information about the original basis, except the space as a whole? Does row reducing to echelon form of the basis not create some sort of useful orthogonal basis?

   I'm absolutely not trying to pick on whoever asked this question, since literally everyone (myself included) has almost certainly done this sort of thing a lot of times at various points, but this is a good example of how mixing up terminology/definitions can make things way more confusing than necessary. Let's start by looking at some of the mathematical phrasing issues in this question:

   (a) A *matrix* can have a column space or a null space, but a basis doesn't have a column space or a null space, since a *a basis isn't a matrix*.

   (b) What is a 'modified identity matrix' supposed to refer to? While we can certainly name a mathematical object whatever we feel like, if we don't *tell the reader* what the thing is that we're naming, then the reader has *no way to know what we're talking about*.

   (c) Similarly to item (a), we *can't* reduce a basis to echelon form. We can only row-reduce a *matrix* and *a basis isn't a matrix*.

   The only reason I bring this up is to point out that, by mixing your terminology/definitions around and by using nonstandard names for things which haven't been defined, it's *almost impossible* for the reader to tell precisely what you're trying to say! Math is tricky enough as-is, so do yourself a favor and write your mathematics with as much precision and detail as possible. If you don't, you're just making an already kind-of-difficult subject even harder for no reason!

   Now, on to the real question that (I think) is being asked here (which, by the way, is a truly excellent question): *Why* do we even care about whether or not a basis is orthogonal or othonormal in the first place?

   The answer to this question comes down to quickly and efficiently dealing with linear transformations. Namely, suppose that $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation and $\mathcal{B} = \left\{v_1, \ldots, v_n\right\}$ is a basis for $\mathbb{R}^n$. Then, since $\mathcal{B}$ is a basis for $\mathbb{R}^n$, every vector $x \in \mathbb{R}^n$ can be written uniquely as $x = c_1 v_1 + \ldots c_n v_n$ for some constants $c_1, \ldots, c_n \in \mathbb{R}$. So, by the linearity of T, we have

$$T(x) = T(c_1 v_1 + \ldots + c_n v_n) = c_1 T(v_1) + \ldots + c_n T(v_n).$$

   That is, by knowing what T does to the n vectors in $\mathcal{B}$, we know what T does to *absolutely every vector in* $\mathbb{R}^n$. This makes computing $T(x)$ a very fast and easy process, since we can figure out what $T(x)$ is just by knowing what the constants $c_1, \ldots, c_n$ are and what the vectors $T(v_1), \ldots, T(v_n)$ are.

   This is where orthogonal bases come in: computing the coefficients $c_1, \ldots, c_n$ is still a very time-consuming task in general (it involves solving a linear system of n equations in n variables). So, anything that makes this faster is definitely useful.

   This is *exactly* what an orthogonal basis lets us do. That is, if $\left\{v_1, \ldots, v_n\right\}$ is an *orthogonal basis* for $\mathbb{R}^n$, then the coefficients $c_1, \ldots, c_n \in \mathbb{R}$ such that $x = c_1 v_1 + \ldots c_n v_n$ are given by

$$c_1 = \frac{x \cdot v_1}{v_1 \cdot v_1}, \quad c_2 = \frac{x \cdot v_2}{v_2 \cdot v_2}, \quad \ldots \quad , c_n = \frac{x \cdot v_n}{v_n \cdot v_n}.$$

   (This is Theorem 5 in Section 6.2 of your textbook)

   Computing a couple of dot products is *much, much faster* than solving a linear system of equations, so an orthogonal basis is definitely a useful thing to have in any practical situation! An *orthonormal basis* just makes this process *even quicker*. That is, if $\left\{v_1, \ldots, v_n\right\}$ is an *orthonormal basis* for $\mathbb{R}^n$, then as $\|v_i\| = 1$ for every $i \in \left\{1, \ldots, n\right\}$, and since $\|v_i\|^2 = v_i \cdot v_i$, we then get that

$$c_1 = x \cdot v_1, \quad c_2 = x \cdot v_2, \quad \ldots \quad , \quad c_n = x \cdot v_n.$$

6. During the review session could you please go over the Gram-Schmidt process as well as identifying an orthogonal basis.
Let's start with identifying an orthogonal basis.

By definition, an *orthogonal set* is a set of vectors $\{v_1, \dots, v_p\}$ such that $v_i \cdot v_j = 0$ for every $i \neq j$, where $i, j \in \{1, \dots, p\}$. By definition, an *orthogonal basis* is just a basis which is also an orthogonal set. So, identifying whether or not a set $\{v_1, \dots, v_p\}$ is an orthogonal basis for a subspace W of $\mathbb{R}^n$ is pretty straightforward: All we need to do is

(a) Check that $\{v_1, \dots, v_p\}$ is a basis for W. That is, check that $\{v_1, \dots, v_p\}$ is linearly independent in $\mathbb{R}^n$ and confirm that $\text{span}\{v_1, \dots, v_p\} = \text{W}$.

(b) Check that $\{v_1, \dots, v_p\}$ is an orthogonal set.

We've dealt with item (a) above for most of the semester, so we'll skip talking about it here. For item (b), all we need to do is check that $v_i \cdot v_j = 0$ for all $i \neq j$, where $i, j \in \{1, \dots, p\}$. All of this is made slightly easier by Theorem 4 of Section 6.2 in your textbook, which states (in a slightly abbreviated form) the following fact:

> **Theorem 2.** *If $\mathcal{S} = \{v_1, \dots, v_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then $\mathcal{S}$ is a basis for the subspace $\text{W} = \text{span}\{v_1, \dots, v_p\}$*

For example, suppose we want to show that the set $\mathcal{S} = \{v_1, v_2, v_3\}$ of vectors $v_1, v_2, v_3 \in \mathbb{R}^3$, where $v_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$, and $v_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$ is an orthogonal basis for $\mathbb{R}^3$. Then, we first show that $\mathcal{S}$ is an orthogonal set. To do this, recall that $x \cdot y = y \cdot x$ (that is, the dot product is symmetric). So, for example, if we show that $v_1 \cdot v_3 = 0$, this also tells us that $v_3 \cdot v_1 = 0$. Therefore, we just need to compute the following dot products:

$$\begin{aligned}
v_1 \cdot v_2 &= (3)(-1) + (1)(2) + (1)(1) = -3 + 2 + 1 = 0 \\
v_1 \cdot v_3 &= (3)(-1/2) + (1)(-2) + (1)(7/2) = -3/2 - 4/2 + 7/2 = 0 \\
v_2 \cdot v_3 &= (-1)(-1/2) + (2)(-2) + (1)(7/2) = 1/2 - 8/2 + 7/2 = 0.
\end{aligned}$$

Since we get zero in each case, we may conclude that $\mathcal{S}$ is an orthogonal set. By the theorem above, since $\mathcal{S}$ is an orthogonal set of nonzero vectors, it is a basis for $\text{span}\{v_1, v_2, v_3\}$. As any three linearly independent vectors in $\mathbb{R}^3$ span $\mathbb{R}^3$, it then follows that $\mathcal{S}$ is an orthogonal basis of $\mathbb{R}^3$.

Now, let's talk about the Gram-Schmidt process. For the reasons explained in Question 5, finding an orthogonal/orthonormal basis often makes things much easier for us in practice. However, *finding* such an orthogonal/orthonormal is generally not a particularly easy or quick task. What the Gram-Schmidt process does is allow us to turn *any* basis **into** an *orthogonal basis* (which, if we then normalize, becomes an orthonormal basis). We do this in what is perhaps the most straightforward way – by *removing* the parts of each vector that are getting in the way of the vector being orthogonal. That is, given any subspace W of $\mathbb{R}^n$, we know that we can write any vector $y \in \mathbb{R}^n$ as $y = \widehat{y} + z$, where $\widehat{y} \in \text{W}$ and $z \in \text{W}^\perp$. So, if we have a basis $\mathcal{B} = \{v_1, \dots, v_p\}$ for a subspace V of $\mathbb{R}^n$, we make a new basis $\mathcal{S} = \{u_1, \dots, u_p\}$ for V by taking $u_1 = v_1$, $u_2 = v_2 - \text{proj}_{\text{span}\{u_1\}}(v_2)$, $u_3 = v_3 - \text{proj}_{\text{span}\{u_1, u_2\}}(v_3)$, and so on, up to $u_p = v_p - \text{proj}_{\text{span}\{u_1, u_2, \dots, u_{p-1}\}}(v_p)$. That is to say, if $\text{W}_i = \text{span}\{u_1, \dots, u_i\}$, then we're simply taking u_{i+1} to be the part of v_{i+1} which lies in $\text{W}_i^\perp$. So, if $v_{i+1} = \widehat{v}_{i+1} + z_{i+1}$, where $\widehat{v}_{i+1} \in \text{W}_i$ and $z_{i+1} \in \text{W}_i^\perp$, we're just setting $u_{i+1} = z_{i+1}$. By doing so, we ensure that u_{i+1} is orthogonal to every vector in the set $\{u_1, \dots, u_i\}$, and by this process, we produce a new, orthogonal basis $\mathcal{S}$ for V from our old, potentially not-orthogonal basis $\mathcal{B}$.

7. To start, how would you prove that any trace of matrix A is equal to the sum of its eigenvalues? Its the extra part of homework 9, question 3.
Question 3 of Homework 9 is:

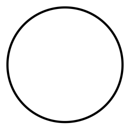
Show that the trace of a diagonalizable matrix A is equal to the sum of the eigenvalues of A.

The intention of this problem, together with Problem 2 in Homework 9, is to help re-enforce that there's nothing inherently difficult about dealing with a concept you might not have seen before. In this case, the concept is the *trace* of a square matrix A, denoted $tr(A)$, which is defined to be the sum of the diagonal entries of the matrix. In problem 2 of Homework 9, you were given the fact that
$$tr(AB) = tr(BA)$$
for any two $n \times n$ matrices A and B. To solve this problem, we just have to use this fact once and the definition of trace once in conjunction with the definition of diagonalizability.

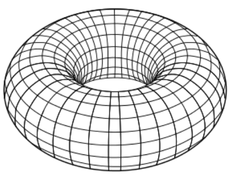
That is, if A is a diagonalizable matrix, then by the definition of diagonalizability, we have $A = PDP^{-1}$ for some invertible matrix P and diagonal matrix D. So, using $tr(AB) = tr(BA)$, we have
$$\begin{aligned} tr(A) &= tr\left(PDP^{-1}\right) \\ &= tr\left((PD)P^{-1}\right) \\ &= tr\left(P^{-1}(PD)\right) \\ &= tr\left(PP^{-1}D\right) = tr\left(D\right). \end{aligned}$$
By the Diagonalization Theorem (Theorem 5 of section 5.3 in your textbook), the diagonal entries of D are the eigenvalues of A. So, as the trace of a matrix is, by definition, the sum of the diagonal entries, the trace of D is the sum of the eigenvalues of A. Since $tr(A) = tr(D)$, we may conclude that the trace of A is the sum of the eigenvalues of A.

8. Is it possible to find the eigenvalues of a torus? It was in Endgame, and Ironman said it with such confidence and sincerity, but I wasn't sure if it was legit or just smart people movie jargon.
Ah! Pop-culture movie-math! What fun!

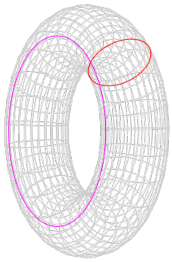
To start, a *torus* is the generic mathematical name for a doughnut' shape in whatever dimension you like. That is, a one-dimensional torus, denoted $\mathbf{T}^1$, is just a circle:



and a two-dimensional torus, denoted $\mathbf{T}^2$, looks like this:



If you look at the two-dimensional torus, notice that it's just a circle dragged around another circle at a 90 degree angle, i.e.



Mathematically, this is a case of what's known as a *product space*. That is, if we defined a notion of multiplication of 'shapes' like this (or, more precisely, a notion of multiplication for topological spaces) in the 'correct way' in some sense, we have $\mathbf{T}^2 = \mathbf{T}^1 \times \mathbf{T}^1$. So, we can make tori of arbitrarily high dimension by taking the n-dimensional torus, called an n-torus and denoted by $\mathbf{T}^n$, to be the product of n 1-dimensional tori, i.e.

$$\mathbf{T}^n = \mathrm{T}^1 \times \ldots \times \mathrm{T}^1.$$

Unfortunately, an n-torus by itself (in whatever dimension) is not a linear transformation, so we can't exactly compute eigenvalues for it. **However**, if we're considering a different object called a 'flat torus' (which is similar, in a certain sense, to the n-torus given above) as a very special type of mathematical object called a *Riemannian manifold* (which is a sort of involved process, where we have to instead define a flat n-torus as the quotient of $\mathbb{R}^n$ by something called a 'lattice of maximal rank'), we **can** find the eigenvalues of a variety of operators that can be implicitly associated to such a Riemannian manifold. In the most likely case (to me, at least), we could (in theory) calculate the eigenvalues of a very important operator called the *Laplace-Beltrami operator* (which, if you've taken a few physics classes, can be seen as a generalization of the Laplacian). So, if we're being generous to the writers of the new Avengers movie, maybe this is what Iron Man was talking about!